

Interaction-Masked Latent Action Models for Object-Aware Manipulation under Visual Distractors

Updated Project Proposal

Core idea: preserve MaskLAM's masked forward-dynamics loss, but extend the FDM from an agent-only target to an interaction-structured foreground model: infer a latent embodiment/action variable, predict the agent transition first, and predict manipulated-object motion through directed attention to the predicted agent transition.

MSc thesis direction proposal

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1 Executive Summary

This proposal examines latent action learning in visually distracting manipulation environments. Latent Action Models (LAMs), such as LAPO, train an inverse dynamics model (IDM) and a forward dynamics model (FDM) on observation-only video so that the IDM infers latent actions and the FDM predicts future observations [2]. Under visual distractors, a full-frame reconstruction loss can pressure the latent action to encode exogenous background motion rather than agent-controlled dynamics. MaskLAM addresses this by applying the FDM reconstruction loss only on agent pixels, using masks obtained from a segmentation foundation model such as SAM2, while otherwise keeping the LAPO-style training pipeline and architecture largely intact [1].

The limitation motivating this proposal is not that MaskLAM simply fails on manipulation. MaskLAM is a strong baseline and achieves high downstream normalized success rate (NSR) on many Distracting Meta-World (DMW) tasks. The sharper gap is that MaskLAM suppresses exogenous background gradients but does not explicitly structure the endogenous foreground interaction. In manipulation, the manipulated object is neither background nor an independently acting agent. It is a task-relevant physical factor whose motion is typically mediated by the robot embodiment.

The proposed method, **Interaction-Masked LAM** (IM-LAM), keeps MaskLAM’s core loss-masking principle and multi-step IDM, but modifies the Stage-1 FDM. The FDM is changed from a monolithic next-frame predictor into an asymmetric, mask-guided interaction predictor. The intended computational path is

$$z_t \longrightarrow \hat{A}_{t+1} \longrightarrow \hat{O}_{t+1}, \quad (1)$$

where z_t is the latent action, A denotes agent or embodiment features, and O denotes manipulated-object features. The object pathway does not receive z_t directly in the main model. Instead, object dynamics are predicted by attending to the current and predicted agent representations. The scene remains visible to the encoder, but it is not an action-conditioned reconstruction target.

The project is evaluated on DMW tasks with residual headroom under MaskLAM, particularly tasks such as **push**, **pick-place**, **door-open**, **window-open**, and **peg-insert-side**. The main metrics remain established LAM metrics: normalized linear action-probe MSE (NMSE) and downstream normalized success rate (NSR). A single auxiliary object-dynamics probe can be added to test whether the object branch captures manipulated-object state or displacement.

2 Context and Background

2.1 Latent action learning

LAMs aim to infer action-like variables from observation-only trajectories. A standard LAPO-style LAM consists of

$$z_t \sim p_{\text{IDM}}(\cdot \mid o_t, o_{t+k}), \quad (2)$$

$$\hat{o}_{t+1} \sim p_{\text{FDM}}(\cdot \mid o_t, z_t), \quad (3)$$

where the IDM infers a latent action and the FDM predicts the next observation. After pretraining on unlabeled video, the learned latent actions can be used to relabel the data and train a latent policy. A small action-labeled subset is then used to learn a decoder from latent actions to executable environment actions [2].

This framework is attractive because it reduces dependence on dense action labels. However, it relies on the assumption that the visual transition is primarily explained by the agent’s action. LAOM and MaskLAM both emphasize that this assumption breaks under action-correlated visual distractors [3, 1].

2.2 MaskLAM and the distractor failure mode

Under a full-frame reconstruction objective, the FDM is asked to predict all visual changes in the next observation. In a distractor setting, the next frame contains both endogenous, action-driven change and exogenous, background or scene change. Since the latent action is the only variable input to the FDM that captures the difference between current and future observations, it can be pressured to encode exogenous dynamics.

MaskLAM’s intervention is to replace the full-frame FDM loss with an agent-masked FDM loss:

$$\mathcal{L}_{\text{FDM}}^{\text{MaskLAM}} = \frac{1}{\|M_{t+1}\|_1} \|M_{t+1} \odot (\hat{o}_{t+1} - o_{t+1})\|_2^2, \quad (4)$$

where M_{t+1} is the target-frame agent mask. This zeroes the reconstruction residual on distractor pixels before it backpropagates into z_t . MaskLAM reports that loss masking is the principal contributor in its ablation study, while the mask input channel and multi-step IDM provide smaller, complementary effects [1].

MaskLAM’s implementation uses a multi-step IDM, a LAPO/LAOM-style IMPALA encoder-decoder world model, stacked observations, and SAM2.1 masks. DMW observations are 128×128 , and the released protocol reports both ground-truth simulator masks and SAM2 masks [1].

2.3 Why manipulation changes the target

MaskLAM’s agent-only target is a strong choice for locomotion-like settings, where the agent pixels are the dominant action-relevant visual region. In manipulation, the manipulated object is not merely background. It is task-relevant, and its transition is usually caused by the embodiment. The endogenous/exogenous split is therefore better described as

$$\text{agent or embodiment} \rightarrow \text{manipulated object}, \quad \text{scene/background} \perp \text{latent action}. \quad (5)$$

The proposed method generalizes MaskLAM’s target from agent pixels to agent-plus-object pixels, but the main contribution is not just adding an object mask. The contribution is to structure the FDM so that the latent action first predicts embodiment motion and object motion is predicted through the predicted embodiment transition.

2.4 Positioning relative to FLAM

FLAM is the closest factored-LAM prior work and should be treated as a central conceptual foil. FLAM starts from the observation that most existing LAM approaches rely on monolithic inverse and forward dynamics models that learn a single latent action to control the entire scene, and therefore struggle in complex environments where multiple entities act simultaneously. FLAM introduces a factored dynamics framework that decomposes the scene into independent factors, each inferring its own latent action and predicting its own next-step factor value [5].

This is closely related but not identical to the present project. FLAM is designed for multi-entity settings where several factors may act simultaneously. In single-agent manipulation, the

manipulated object is usually not an independently acting factor; it is a passive or tool-like factor whose motion is mediated by the robot embodiment. Giving the object a full independent latent-action degree of freedom may allow the model to explain object motion through a separate object-action channel instead of forcing it through the predicted agent transition.

IM-LAM therefore adopts an asymmetric factorization:

FLAM: independent factors, each with its own latent action, designed for complex multi-entity settings where multiple entities act simultaneously.

IM-LAM: one embodiment/action latent predicts the agent transition; object dynamics are predicted autoregressively from that predicted transition.

2.5 Positioning relative to OCL

Object-Centric Latent Action Learning (OCL) uses VideoSAUR slots and a LAPO-style pipeline to improve latent action learning under distractors [4]. However, slot-based methods may produce functional rather than semantic decompositions, and the MaskLAM paper reports instability of LAPO-Slots when the relevant agent identity is not consistently assigned to a fixed slot index across samples [1]. This motivates the use of explicit segmentation masks and a mask-guided FDM rather than relying on slot identity stability.

3 Problem Statement

Existing label-free LAM methods under distractors leave an interaction-structure gap:

1. **Agent-only masked methods**, such as MaskLAM, successfully suppress background gradients but do not explicitly model manipulated objects as downstream effects of agent motion.
2. **Symmetric factored LAMs**, such as FLAM, give every factor its own latent action, which is appropriate for multiple simultaneously acting entities but less aligned with passive manipulated objects in single-agent manipulation.
3. **Object-centric slot methods**, such as OCL/LAPO-Slots, can improve representation quality but may suffer from unstable or non-semantic slot binding.

Gap: There is no LAM method that preserves MaskLAM’s robust loss-masking mechanism while explicitly imposing the single-agent manipulation prior that latent actions first explain embodiment motion and object motion is predicted as a consequence of the predicted embodiment transition.

4 Research Aim, Question, and Hypothesis

4.1 Aim

The aim is to develop and evaluate an interaction-structured extension of MaskLAM for distracting manipulation tasks, in which the latent action is treated as an embodiment-action latent and the manipulated object’s transition is predicted through a directed agent-to-object FDM.

4.2 Research question

Can an interaction-structured masked LAM improve latent action quality, object-dynamics representation, and downstream manipulation performance under visual distractors by reconstructing both agent and object pixels while enforcing a directed agent-to-object dynamics pathway in the FDM?

4.3 Hypothesis

In distracting manipulation environments, agent-only MaskLAM suppresses exogenous background pressure but does not explicitly model task-relevant object consequences. A directed FDM that predicts the agent transition from the latent action and then predicts the object transition from the predicted agent transition will produce more useful latent actions and object-dynamics representations than an agent-only or foreground-only masked LAM on tasks where object/contact dynamics leave measurable headroom.

This hypothesis is not a claim of causal discovery. It is an architectural causal prior: embodiment acts, object responds, and scene/background remains exogenous.

4.4 Objectives

1. Reproduce or approximate a MaskLAM-style baseline on a subset of Distracting Meta-World.
2. Extend mask preprocessing to obtain both agent and manipulated-object masks.
3. Implement a Foreground-MaskLAM baseline that reconstructs agent and object pixels without directed interaction modeling.
4. Implement Interaction-Masked LAM with a directed agent-to-object FDM.
5. Evaluate latent action quality via normalized linear action-probe MSE.
6. Evaluate downstream policy performance via DMW normalized success rate under limited action labels.
7. Add one lightweight object-dynamics diagnostic: a linear probe from the object branch representation to object pose, object mask centroid, or object displacement.

5 Proposed Model: Interaction-Masked LAM

5.1 High-level design

The model keeps the three-stage LAM pipeline used by LAPO and MaskLAM:

1. Stage 1: train IDM and FDM on observation-only video.
2. Stage 2: relabel the dataset with latent actions and train a latent policy.
3. Stage 3: train a small action decoder on a limited set of ground-truth action labels.

The proposed change is in Stage 1: the FDM is changed from a monolithic next-frame predictor into a directed interaction predictor. The IDM remains close to MaskLAM’s multi-step IDM.

5.2 Notation

Let:

- $o_t \in \mathbb{R}^{H \times W \times 3}$: RGB observation at time t , with DMW using 128×128 observations in MaskLAM’s protocol.
- M_t^A : agent or embodiment mask at time t .
- M_t^O : manipulated-object mask at time t .
- $z_t \in \mathbb{R}^{d_z}$: latent action inferred by the IDM.
- B_t : FDM bottleneck feature map, the low-resolution spatial tensor between the FDM encoder and decoder.

5.3 Mask generation

The preferred mask source is SAM2/SAM2.1 video segmentation, initialized with a first-frame box or point prompt for the agent and manipulated object. SAM2 is designed for promptable image and video segmentation with streaming memory and supports interactive video object propagation [6]. MaskLAM uses SAM2.1-hiera-tiny to precompute agent masks and reports that SAM masks track ground-truth mask performance closely in DMW [1].

If first-frame boxes are unreliable for object masks, an open-vocabulary detector such as Grounding DINO can be used only as an initialization tool. The main experimental comparison should keep the mask source fixed across MaskLAM-agent, Foreground-MaskLAM, and Interaction-Masked LAM.

5.4 IDM

The IDM remains close to MaskLAM. MaskLAM uses a multi-step IDM with three current frames and three future frames, where the future offset k is sampled during training. In MaskLAM, the IDM consumes current and future RGB frame stacks augmented with the agent mask channel, and the latent action dimension is typically $d_z = 128$ [1].

For the proposed method, the main IDM follows this MaskLAM convention:

$$z_t \sim p_{\text{IDM}} \left(\cdot \mid o_{t:t+2}, o_{t+k:t+k+2}, M_{t:t+2}^A, M_{t+k:t+k+2}^A \right). \quad (6)$$

The object mask is not part of the IDM input. This keeps z_t interpretable as an embodiment-action latent. Object dynamics are introduced in the FDM as a downstream consequence of the predicted embodiment transition. The current proposal does not include a separate IDM mask-channel ablation; the IDM is kept close to MaskLAM so that the contribution is localized to the FDM.

5.5 FDM bottleneck

The FDM bottleneck is the low-resolution spatial feature map produced by the FDM encoder before the decoder upsamples back to RGB. MaskLAM’s FDM is implemented as an IMPALA-style encoder-decoder world model: the current frame stack and mask are encoded, z_t is projected to the bottleneck spatial shape and concatenated channel-wise, and a symmetric decoder reconstructs the next RGB observation [1].

For IM-LAM, the FDM receives the current RGB stack together with agent and object mask information. The masks are used both as input hints, following MaskLAM’s mask-conditioned FDM convention, and as routing signals for the bottleneck interaction module. For DMW, the input resolution is 128×128 . With three downsampling blocks, the bottleneck is approximately a 16×16 spatial feature grid. This is the natural place to insert the proposed interaction module: it is spatial enough for masks to localize agent/object regions, compact enough for attention, and already the location where z_t is injected into the FDM.

5.6 Mask2Former-inspired masked cross-attention

Mask2Former introduced masked attention, where cross-attention is constrained to localized mask regions so that object queries extract localized features rather than attending globally [7]. The proposed method borrows this principle without importing the full Mask2Former segmentation architecture.

At the FDM bottleneck, the current-frame masks are downsampled to bottleneck resolution and used as soft attention biases:

$$W_t^A = \text{Downsample}(M_t^A), \quad (7)$$

$$W_t^O = \text{Downsample}(M_t^O). \quad (8)$$

Learned agent and object queries extract entity tokens from the bottleneck feature map:

$$A_t = \text{MaskedCrossAttn}(Q_A, B_t; W_t^A), \quad (9)$$

$$O_t = \text{MaskedCrossAttn}(Q_O, B_t; W_t^O). \quad (10)$$

The masks are used as soft localization priors, not as hard image crops. The full image remains visible to the encoder, consistent with MaskLAM’s finding that input masking can remove useful interaction context while loss masking removes exogenous target pressure [1].

5.7 Directed agent-to-object FDM

The proposed FDM has two directed stages.

5.7.1 Stage 1: agent dynamics

The latent action updates the agent pathway using MaskLAM-style FDM machinery:

$$\hat{A}_{t+1} = F_A(A_t, z_t). \quad (11)$$

The agent update can be implemented at the bottleneck using residual convolutional blocks and the same projected z_t injection used by MaskLAM’s FDM. This keeps the agent branch close to the original architecture.

5.7.2 Stage 2: object dynamics

The object pathway is updated through directed cross-attention to the current and predicted agent representations:

$$\hat{O}_{t+1} = F_O \left(O_t, \text{CrossAttn} \left(Q = O_t, K, V = [A_t, \hat{A}_{t+1}] \right) \right). \quad (12)$$

The object branch does not receive z_t directly. This enforces the intended information path:

$$z_t \rightarrow \hat{A}_{t+1} \rightarrow \hat{O}_{t+1}. \quad (13)$$

The transition difference $\hat{A}_{t+1} - A_t$ is not passed as an additional key/value in the main model, because the object branch already attends to both A_t and $\hat{A}_{t+1}$. If useful in implementation, temporal/type embeddings can distinguish current-agent tokens from predicted-next-agent tokens.

5.8 Write-back and shared decoder

The predicted object tokens are written back into the FDM bottleneck using object-mask-biased routing. The final bottleneck prediction is:

$$\hat{B}_{t+1} = B_t + \Delta B_A + \Delta B_O, \quad (14)$$

where ΔB_A and ΔB_O are soft spatial updates localized by dilated current-frame agent and object masks.

A single shared MaskLAM-style decoder then predicts the next RGB observation:

$$\hat{o}_{t+1} = D(\hat{B}_{t+1}). \quad (15)$$

This is preferred over separate agent/object decoders because it preserves the standard world-model formulation: one FDM predicts one next observation, and the loss decides which pixels supervise the prediction.

5.9 Optional object-residual latent

The main model assumes that object motion is predictable from the current object state and the predicted agent transition. This is a strong but appropriate prior for quasi-static pushing and many rigid-contact manipulation tasks. In some precision-contact settings, however, object motion may be underdetermined by the visible agent transition because of slip, release, occlusion, or contact ambiguity.

As a diagnostic extension, the model may include a small, heavily bottlenecked object-residual latent r_t^O :

$$\hat{O}_{t+1} = F_O(O_t, A_t, \hat{A}_{t+1}, r_t^O), \quad (16)$$

with r_t^O constrained by small dimensionality and an L_2 or KL-style regularizer. This is not the main action representation and is not intended to become a full FLAM-style object action. It is a residual channel for contact ambiguity. If it activates mainly on **peg-insert-side** and remains near-zero on simpler pushing or opening tasks, that would support the interpretation that the directed prior is correct by default but needs limited residual capacity for highly ambiguous contact.

5.10 Scene handling

The scene is treated as visible but exogenous context. It is visible to the encoder through the full RGB input, but it is not an action-conditioned target:

- no direct scene reconstruction loss in the main objective;

- no path from z_t to a scene-prediction branch;
- no scene input to the IDM beyond whatever context the full-image encoder naturally contains.

This follows the MaskLAM principle that exogenous background variation should not provide gradients to the latent action.

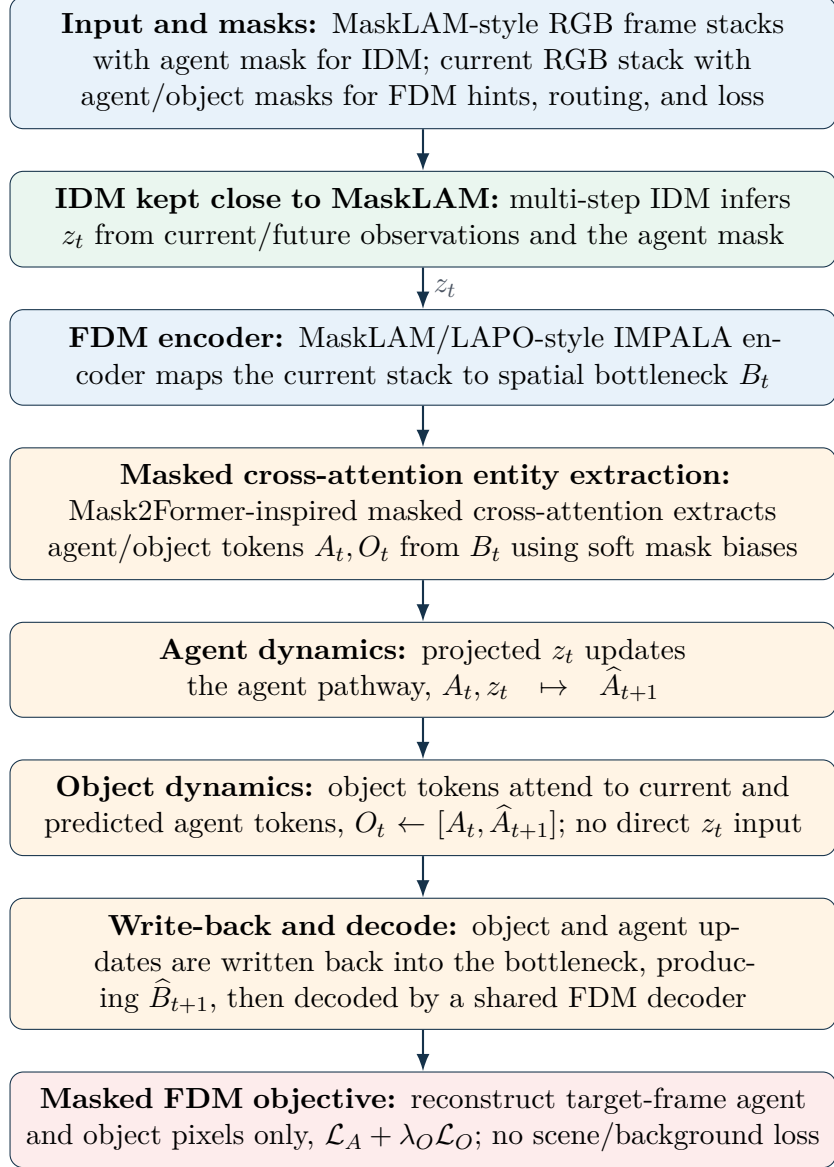


Figure 1: Proposed Interaction-Masked LAM architecture. The design preserves MaskLAM’s multi-step IDM and encoder-decoder backbone while adding a directed agent-to-object interaction module at the FDM bottleneck.

6 Training Objective

MaskLAM uses an area-normalized masked reconstruction loss on target-frame agent pixels. The proposed method generalizes this to two endogenous target regions: the agent/embodiment and the manipulated object.

Define:

$$\mathcal{L}_A = \frac{1}{\|M_{t+1}^A\|_1} \left\| M_{t+1}^A \odot (\hat{o}_{t+1} - o_{t+1}) \right\|_2^2, \quad (17)$$

$$\mathcal{L}_O = \frac{1}{\|M_{t+1}^O\|_1} \left\| M_{t+1}^O \odot (\hat{o}_{t+1} - o_{t+1}) \right\|_2^2. \quad (18)$$

The main FDM loss is:

$$\mathcal{L}_{\text{IM-LAM}} = \mathcal{L}_A + \lambda_O \mathcal{L}_O. \quad (19)$$

The initial setting is $\lambda_O = 1.0$, because both terms are mask-area normalized. There is no scene/background reconstruction term in the main objective. This preserves the key MaskLAM gradient property: exogenous pixels do not shape the latent action. The difference is that the manipulated object is treated as an endogenous target, not as background.

For the optional object-residual variant, an additional small penalty may be used:

$$\mathcal{L}_{\text{res}} = \beta \|r_t^O\|_2^2 \quad \text{or} \quad \beta D_{\text{KL}}(q(r_t^O) \| \mathcal{N}(0, I)), \quad (20)$$

with a small β chosen to keep the residual latent from becoming a second unconstrained object-action channel.

7 Experiments and Evaluation

7.1 Benchmark and task subset

The primary benchmark is Distracting Meta-World (DMW), following the MaskLAM/OCL/LAOM distractor protocol. MaskLAM evaluates DMW at 128×128 pixels with dynamic DAVIS background distractors and reports normalized success rate over Meta-World tasks [1]. DMW is preferable to pure locomotion for this proposal because the research hypothesis concerns manipulated-object dynamics.

The evaluation should report per-task results rather than relying only on the MT10 average. Several DMW tasks are already near-saturated under MaskLAM, so the primary focus should be tasks with residual NMSE or NSR headroom and meaningful object/interaction structure:

- **push**: quasi-static object interaction; a natural fit for the directed embodiment-to-object prior.
- **pick-place**: object-relevant and still leaves meaningful downstream headroom.
- **door-open** and **window-open**: object/fixture interaction with nontrivial residual probe and success gaps.
- **peg-insert-side**: largest headroom and most contact-sensitive; also the highest-risk task for a deterministic object pathway.

A full MT10 evaluation is a stretch goal. If run, it should be treated as secondary context because saturated tasks may obscure the effect of interaction modeling.

7.2 Baselines

The minimal comparison set is:

Method	Purpose
<i>LAPO</i>	Standard unmasked LAM baseline.
MaskLAM-agent	Strong agent-only masked baseline. This tests whether agent-only masking is sufficient.
Foreground-MaskLAM	Same MaskLAM-style FDM, but loss on agent $\cup$ object pixels. This isolates the effect of adding object pixels to the target.
Interaction-Masked LAM	Proposed model with directed agent-to-object FDM.
Interaction-Masked LAM + object residual	Diagnostic variant for contact ambiguity; expected to matter most on precision-contact tasks such as peg-insert-side .

LAOM-Labels can be included as an upper bound if implementation time permits, because it uses privileged ground-truth action supervision during LAM training [3]. OCL/LAPO-Slots and FLAM are essential related work, but not required core baselines unless code, data, and time make them feasible.

7.3 Metrics

Use established metrics from MaskLAM and LAOM:

- **Normalized linear action-probe MSE (NMSE):** train a linear regressor from frozen latent actions to clipped ground-truth actions. Lower is better.
- **Normalized success rate (NSR):** evaluate the decoded policy on DMW rollouts and normalize by expert success.
- **Object-dynamics probe:** train a frozen linear probe from the object-branch representation to an object-state target such as object pose, object mask centroid, or object displacement. This is a secondary diagnostic to test whether the proposed object pathway actually captures manipulated-object dynamics.
- (Label-budget sweep): evaluate Stage 3 decoder performance using a small set of ground-truth action labels, e.g. 2k, 16k, and 128k labels, if time permits.

Full DCI-style disentanglement analysis is optional and not part of the main plan. Agent and object are intentionally coupled in manipulation, so a simple object-dynamics probe is more directly aligned with the proposed contribution.

7.4 Expected result pattern

The expected result is task-dependent. The strongest improvement should appear on manipulation tasks where object state affects success and where MaskLAM-agent leaves residual NMSE or NSR headroom.

A positive result would entail:

$$\text{Interaction-Masked LAM} > \text{Foreground-MaskLAM} \geq \text{MaskLAM-agent} > \text{LAPO} \quad (21)$$

for downstream NSR on object-centric DMW tasks. The most important comparison is against Foreground-MaskLAM, because it determines whether directed agent-to-object dynamics add value beyond simply reconstructing the manipulated object.

If the residual object latent improves only on `peg-insert-side` while remaining unused or unhelpful on `push` and `door-open`, that would support the interpretation that most object motion is mediated by agent transition, but contact-ambiguous tasks may require limited residual capacity.

8 Feasibility

The proposal is designed to be close to MaskLAM’s computational regime. MaskLAM reports that DMW observations are 128×128 , that the method uses an IMPALA encoder/decoder, and that DMW MaskLAM training is feasible on a single A100-class GPU with peak memory below 24 GB and single-task wall-clock on the order of hours rather than days [1]. The proposed method adds a small bottleneck attention module and an object mask stream for loss/routing, but does not train a large pretrained visual encoder or a DINO-WM-style planner.

The main engineering cost is mask generation and integration. MaskLAM reports that SAM2.1 masks can be precomputed and stored alongside the dataset, and that ground-truth and SAM masks can both be used for evaluation [1]. For this thesis, the first implementation can use a small task subset and precomputed masks. Agent masks can follow MaskLAM’s first-frame prompt procedure. Object masks require task-specific prompts or boxes, but SAM2 video propagation makes this tractable [6].

9 Risks and Mitigations

Risk	Mitigation
Object masks are noisy or unstable.	Start with tasks where the manipulated object is visually clear; compare SAM masks to available simulator masks if provided; use mask dilation/erosion robustness checks following MaskLAM.
MaskLAM-agent already saturates many DMW tasks.	Focus primary evaluation on tasks with residual NMSE/NSR headroom: <code>peg-insert-side</code> , <code>pick-place</code> , <code>push</code> , <code>door-open</code> , <code>window-open</code> , and <code>reach</code> . Report MT10 only as secondary context.
Foreground-MaskLAM matches the proposed model.	Interpret this as evidence that object target selection is sufficient and directed FDM structure is not needed on the selected tasks; analyze task dependence and object-dynamics probe results.
Strict object pathway underfits contact ambiguity.	Add the small residual object-latent diagnostic variant and report whether residual usage is task-specific.
Attention module overfits or destabilizes training.	Keep the module at bottleneck resolution with few queries and one or two attention layers; preserve MaskLAM hyperparameters elsewhere.
Action-probe NMSE does not improve despite better success.	Treat downstream DMW success and object-dynamics probe as co-evidence. The latent action is intentionally embodiment-centered, so action-probe gains may be smaller than policy or object-prediction gains.

10 Proposed Timeline

Period	Milestone
Weeks 1–2	Obtain/reproduce MaskLAM or LAOM/LAPO baseline on one DMW task; verify masks and Stage 1–3 pipeline.
Weeks 3–4	Add object masks and implement Foreground-MaskLAM baseline.
Weeks 5–6	Implement Interaction-Masked FDM bottleneck module and train on one object-centric task.
Weeks 7–8	Run main experiments on 2–4 DMW tasks, with three seeds if feasible.
Weeks 9–10	Residual object-latent diagnostic, object-dynamics probe, label-budget sweep if feasible, visualizations, and write-up.

11 Conclusion

This project proposes a focused extension of MaskLAM that addresses a manipulation-specific gap. MaskLAM suppresses exogenous distractor gradients by applying the FDM loss only on agent pixels. FLAM shows that factorized latent action models are useful when multiple entities act simultaneously, but its symmetric factorization assigns a latent action to each factor. IM-LAM instead targets single-agent manipulation: one embodiment/action latent predicts the agent transition, and object dynamics are predicted autoregressively from that transition.

The contribution is therefore not merely adding another mask. The contribution is a directed information path:

$$\text{latent action} \rightarrow \text{predicted embodiment transition} \rightarrow \text{predicted object transition}, \quad (22)$$

with scene/background removed from the action-conditioned reconstruction target. This is a concrete, computationally feasible implementation of embodiment-object-scene factorization in a LAM setting.

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